

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Differentiate between Von-Neumann and Harvard architecture. 5
- b) Explain in detail the flags of 8085 microprocessor. 5
- c) Write an 8085 ALP to find out the smallest number in an array available from memory location starting from 2500H to 2510H and Store the result in the register C. 5
- a) What happens when HLT instruction is executed in processor? 4
- b) Write an 8085 ALP to multiply the number 37H and 42H and keep the result in memory. 5
- c) Design a circuit to interface a 2 KB EPROM to 8085 microprocessor .The starting address being 1000H. 6
- a) Draw the timing diagrams for the instruction STA 400FH. What does quality factor mean in microprocessor? 7+1
- b) What is control word? Write a program to take input from 8 switches connected to port B of 8255A and display the status of switches of 8, LED connected to port A of 8255A (for mode 0). 7
- a) With the help of the register organization diagram explain the types of registers in 8086 microprocessor. 7
- b) What is interrupt and interrupt service routine? Explain polled, vectored interrupt and non-maskable interrupt with examples. 8
- a) Define "Assembler Directives". Explain the following assembler directives along with their format: The Memory Model Definition, The PROC directive, The .STARTUP directive, The .EXIT directive and The END directive. 7
- b) Define Procedure and Macros and explain the addressing mode used in 8086 microprocessor with examples. 8

6. a) Write an 8086 ALP for MASM to display the text "POKHARAUNIVERSITY" in reverse order. 8
- b) Two computers are to be connected with each other via modems. Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standards? 7
7. Write short notes on: (Any two) 2*5
- a) RS 422A
- b) SRAM Vs DRAM
- c) 80386 microprocessor and its type